

collaborative PP-Module for DBMS in the Cloud

Version 0.3, 2026-02-23

Acknowledgements

This collaborative Protection Profile Module (PP-Module) was developed by the Database Management Systems international Technical Community (iTC) also known as DBMS-iTC with representatives from industry, Government agencies, Common Criteria Test Laboratories, and members of academia.

INDUSTRY

Vendors

COMMON CRITERIA TESTING LABORATORIES

ITSEF Name

GOVERNMENT AGENCIES

Government Agency Names

Revision History

Table 1. Revision history

Version	Date	Description
0.1	2025-03-12	Initial Release for iTC Review
0.2	2026-01-25	CC:2022 Compliance Update - Improved SPD with enhanced threat descriptions and new threats (T.SECRETS_EXPOSURE, T.NETWORK_INTERCEPTION), improved assumptions, added organizational security policies (P.AUDIT_RETENTION, P.ACCESS_CONTROL_POLICY), added new objective (O.CONFIGURATION_PROTECTION), improved OE objectives with conditional triggers, complete SFR reformatting with CC:2022-compliant element numbering, explicit dependency resolution, updated conformance claims to CC:2022, added PP-Module Base section

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Preface

This PP-Module, the **DBMS in the Cloud Module**, extends the collaborative Protection Profile for Database Management Systems (cPP_DBMS) to support secure deployment of DBMS products in cloud environments. It is intended to address lift-and-shift scenarios where a traditional DBMS is deployed by a tenant on top of infrastructure- or platform-as-a-service (IaaS/PaaS) offerings in public or private clouds.

This module is aligned with the architectural concepts defined by the Common Criteria in the Cloud Technical Community (CCiTC), as described in the "Guidance for Cloud Evaluations" v1.1 publication. In particular, this module assumes the presence of a **Trusted Platform** — that is, a set of services and infrastructure provided by the cloud provider that enforce essential baseline security properties such as isolation, resource integrity, and identity attestation. The Trusted Platform is considered part of the TOE's operational environment and is not within the scope of evaluation.

The DBMS in the Cloud Module is focused on scenarios where:

- The TOE is functionally equivalent to its on-premises version and is not operated or managed by the cloud provider.
- The TOE is installed, configured, and administered by the customer (tenant) or a delegated administrator.
- The TOE may be deployed using containers, virtual machine images, or native installation on cloud-provisioned compute resources.
- The operational environment provides cloud-native identity, audit, and configuration management services with which the TOE may integrate.

This PP-Module is **not** intended for Database-as-a-Service (DBaaS) offerings, where the cloud provider operates and manages the DBMS on behalf of tenants. A separate PP-Module will define requirements for DBaaS environments and is expected to require conformance to the cPP_DBMS, the DBMS Crypto Module, and additional DBaaS-specific interface controls via a PP-Configuration.

This Cloud Module introduces SFRs to support:

- Integration with cloud-native identity and access management (IAM) systems
- Secure consumption of cloud-based secrets and configuration data
- Export of audit logs to cloud-native observability and logging platforms
- Monitoring of cloud deployment integrity and alerting on failure or misconfiguration
- (Via selection) Use of cryptographic mechanisms for data-in-transit and data-at-rest protection, as defined in the DBMS Crypto Module

This PP-Module does not stand alone. It must be claimed in conjunction with:

- The cPP_DBMS (Base PP)
- Optionally, the DBMS Crypto Module when cryptographic protections are selected

By layering this PP-Module with the cPP_DBMS and the Crypto Module, vendors and evaluators can support secure, standards-based assurance for DBMS products deployed in modern cloud environments without conflating service provider responsibilities with product-level evaluation scope.

Objectives of Document

This document presents the Common Criteria (CC) collaborative Protection Profile Module (PP-Module) to express the security functional requirements (SFRs) and security assurance requirements (SARs) for Database Management Systems deployed in cloud environments. The Evaluation activities that specify the actions the evaluator performs to determine if a product satisfies the SFRs captured within this PP-Module, are described in [\[SD\]](#).

Scope of Document

The scope of the PP-Module within the development and evaluation process is described in the Common Criteria for Information Technology Security Evaluation, CC:2022. In particular, a PP-Module defines the IT security requirements of a generic type of TOE and specifies the functional security measures to be offered by that TOE to meet stated requirements [\[\[CC1\]](#), Section B.14].

Intended Readership

The target audiences of this PP-Module are developers, CC consumers, system integrators, evaluators and schemes.

Although the PP-Module and SD may contain minor editorial errors, the PP-Module is recognized as living document and the iTC is dedicated to ongoing updates and revisions. Please report any issues to the DBMS-iTC.

Related Documents

- [\[CC1\]](#) Common Criteria for Information Technology Security Evaluation, Part 1: Introduction and General Model, CC:2022, Version 2022, Revision 1.
- [\[CC2\]](#) Common Criteria for Information Technology Security Evaluation, Part 2: Security Functional Components, CC:2022, Version 2022, Revision 1.
- [\[CC3\]](#) Common Criteria for Information Technology Security Evaluation, Part 3: Security Assurance Components, CC:2022, Version 2022, Revision 1.
- [\[CC4\]](#) Common Criteria for Information Technology Security Evaluation, Part 4: Framework for the Specification of Evaluation Methods and Activities, CC:2022.
- [\[CC5\]](#) Common Criteria for Information Technology Security Evaluation, Part 5: Pre-defined Packages of Security Requirements, CC:2022.
- [\[CEM\]](#) Common Methodology for Information Technology Security Evaluation, CEM:2022.
- [\[SD\]](#) Supporting Document - Evaluation Activities for DBMS in the Cloud Module.
- [\[cPP_DBMS\]](#) collaborative Protection Profile for Database Management Systems.

- [DBMS_MOD_CRYPT0] PP-Module for DBMS Cryptographic Functions.

For more information, see the [Common Criteria Portal](#).

Chapter 1. PP-Module Introduction

1.1. PP-Module Reference Identification

This section provides the formal identification of this PP-Module per APE_INT requirements.

Table 2. PP-Module Identification

Attribute	Value
PP-Module Title	collaborative PP-Module for DBMS in the Cloud
PP-Module Short Name	DBMS_Cloud_MOD
PP-Module Version	0.3
PP-Module Publication Date	2026-02-23
PP-Module Sponsor	Database Management Systems international Technical Community (DBMS-ITC)
CC Version	CC:2022
PP-Module Keywords	Database, DBMS, Cloud, IaaS, PaaS, Lift-and-Shift

Chapter 2. PP-Module Base

This section specifies the Base PP(s) that must be used in conjunction with this PP-Module and defines the allowed PP-Configurations, per CC:2022 requirements for PP-Module conformance claims.

2.1. Required Modules

This PP-Module requires the following PP-Module to be claimed in conjunction with the Base PP:

- **PP-Module for DBMS Cryptographic Functions (DBMS_MOD_CRYPTO)**
[[DBMS_MOD_CRYPTO]]

Rationale: Deploying a DBMS in a cloud environment using a "lift-and-shift" model exposes management interfaces, audit channels, and IAM integration traffic to shared cloud networks. Data-in-Transit protection (TLS/HTTPS) is therefore a mandatory baseline security property. The DBMS Crypto Module provides the necessary Security Functional Requirements (SFRs) for this protection.

Note: While the Crypto Module is mandatory, the specific implementation of Data-at-Rest Encryption (D@RE) remains flexible. The ST author may select either "TOE-Implemented" encryption or "Leverage Platform Encryption" within the Crypto Module's FDP_DAR_EXT.1 requirement.

2.2. Allowed PP-Configurations

The only allowed PP-Configuration for this module is:

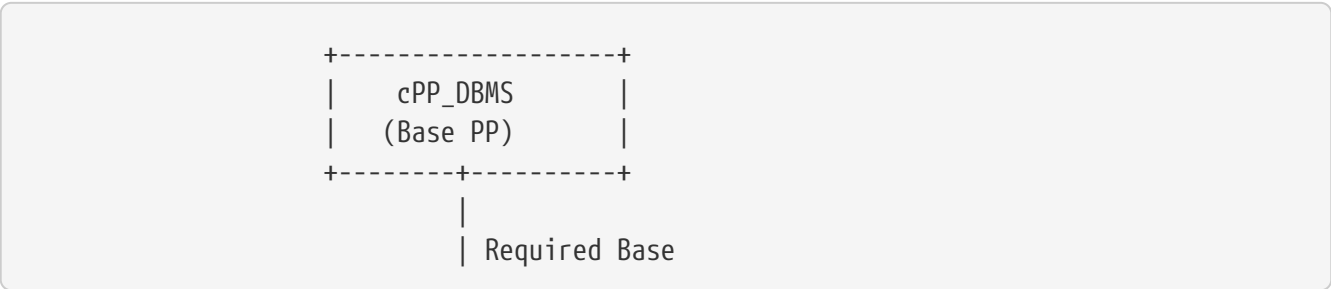
1. **cPP_DBMS + DBMS Cloud Module + DBMS Cryptographic Module**

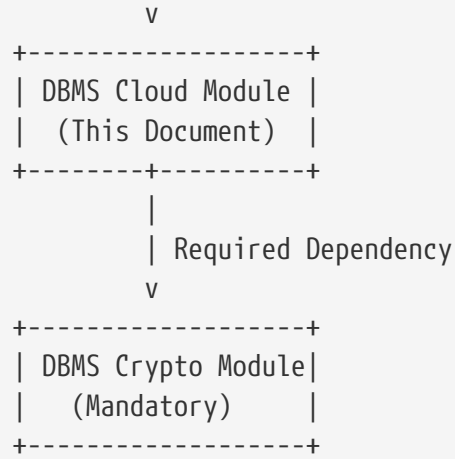
This configuration is required for all evaluations. It ensures that the TOE provides:

- **Data-in-Transit Protection:** Mandatory for all cloud deployments to protect management traffic and service integrations (provided by the Crypto Module).
- **Data-at-Rest Protection:** Flexible implementation (TOE-managed or Platform-managed) selected within the Crypto Module.

Any PP-Configuration claiming this PP-Module that does not include the DBMS Cryptographic Module is not valid.

2.3. PP-Module Relationship Diagram





2.4. Future Modules

The following PP-Modules are planned for future development:

- **DBMS DBaaS Module** [DBMS_MOD_DBAAS] - Will extend scope for Database-as-a-Service deployments where the cloud provider operates and manages the DBMS on behalf of tenants. This module is expected to require conformance to the cPP_DBMS, the DBMS Crypto Module, and additional DBaaS-specific interface controls.

Chapter 3. TOE Overview

This PP-Module applies to Database Management Systems (DBMS) deployed in cloud environments. The Target of Evaluation (TOE) is the DBMS software itself when deployed by the customer or tenant on infrastructure-as-a-service (IaaS) or platform-as-a-service (PaaS) offerings provided by public or private cloud providers.

3.1. TOE Type

The TOE type addressed by this PP-Module is: Database Management Systems deployed in cloud environments.

3.2. TOE Description

The TOE is functionally equivalent to a traditional, on-premises DBMS. It may be delivered as a binary package, virtual machine image, or containerized application and is deployed and administered by the end user, not by the cloud service provider. This PP-Module does **not** apply to database-as-a-service (DBaaS) offerings operated by the cloud provider; those deployments are out of scope and will be addressed in a separate PP-Module.

The TOE is hosted on a **Trusted Platform**, which encompasses the cloud infrastructure and services relied upon by the TOE to enforce its security policies. The Trusted Platform includes:

- Compute, storage, and network resources provisioned via the cloud control plane
- Orchestration and deployment systems (e.g., Kubernetes, Terraform)
- Cloud-native identity and access management (IAM) systems
- Secret management, logging, and monitoring services

The TOE assumes that the Trusted Platform enforces proper isolation, provides resource availability, and supplies accurate identity and policy assertions. These assumptions are captured in the Operational Environment objectives defined by this module.

3.3. TOE Scope

The TOE includes:

- The DBMS application software and any tightly integrated subsystems
- Interfaces used to configure and manage security-relevant behavior (e.g., authentication settings, audit policy)
- Internal communications between distributed DBMS components (e.g., nodes in a cluster)
- Any embedded runtime used to bootstrap or orchestrate the DBMS (e.g., scripts, sidecar services)

The TOE explicitly excludes:

- The Trusted Platform, including the hypervisor, cloud control plane, and underlying hardware
- Cloud-native IAM services, monitoring systems, and key vaults (though the TOE may integrate with them)
- Third-party cryptographic modules unless explicitly claimed via the DBMS Crypto Module

When selections are made to claim cryptographic protections (e.g., TLS, data-at-rest encryption), this module expects the TOE to include the DBMS Crypto Module as defined in its associated PP-Configuration.

3.4. Major Security Features

The TOE provides the following major security features:

- **Cloud Identity Integration:** Integration with external cloud-native identity and access management (IAM) systems for user authentication and role mapping.
- **Cloud Audit Integration:** Generation of audit records with support for export to cloud-native or tenant-managed audit collection systems.
- **Elastic Environment Resilience:** Maintenance of security properties in ephemeral, scaled, or orchestrated cloud deployments.
- **Deployment Integrity:** Verification of runtime state and deployment artifacts when such capabilities are implemented.
- **Secure Configuration Management:** Protection of TSF data stored in cloud-native configuration and secret management systems.

Chapter 4. TOE Use Case

This section describes a representative deployment scenario for the TOE, illustrating the practical necessity of selecting cryptographic protections defined in the DBMS Crypto Module. This scenario is provided to explicitly justify the enforcement of Data-in-Transit (DIT) and Data-at-Rest Encryption (D@RE) selections through the PP-Configuration, facilitating NIAP approval for inclusion on the Product Compliant List (PCL).

4.1. Cloud Deployment Scenario (Public Cloud IaaS)

The TOE is deployed by an enterprise customer using a "lift-and-shift" strategy onto a public cloud provider's infrastructure-as-a-service (IaaS) offering. The deployment consists of:

- A DBMS installed on cloud-provisioned virtual machine instances or containers.
- Persistent database storage provisioned using cloud-managed storage volumes.
- Administrative and user interactions with the DBMS conducted remotely over untrusted public networks.

In this scenario, the following cryptographic protections are essential:

- **Data-in-Transit Protection (DIT):** Administrative interfaces and API endpoints are exposed externally for remote management and access. Without cryptographic protections, administrative credentials, queries, and results transmitted across the network are vulnerable to interception or manipulation. **Required selection:** TLS or HTTPS as specified in [FDP_DIT_EXT.1](#) and defined in the DBMS Crypto Module.
- **Data-at-Rest Encryption (D@RE):** Sensitive or regulated data (e.g., customer PII, financial data, or intellectual property) is stored persistently within cloud-managed storage volumes. Without encryption, data confidentiality is at risk if underlying storage is compromised or accessed by unauthorized parties. **Required selection:** Encryption of persistent data storage as specified in [FDP_DAR_EXT.1](#) and defined in the DBMS Crypto Module.

This use case explicitly justifies the necessity of selecting and implementing cryptographic requirements defined in the DBMS Crypto Module. Thus, any TOE claiming conformance to this PP-Module under these deployment conditions must explicitly select these cryptographic protections within its PP-Configuration.

Chapter 5. CC Conformance Claims

This section describes the conformance claims for this PP-Module per APE_CCL requirements defined in CC:2022.

5.1. Common Criteria Conformance

This PP-Module claims conformance to the Common Criteria for Information Technology Security Evaluation, CC:2022, as follows:

- **CC Part 1 Conformance:** This PP-Module is consistent with CC:2022 Part 1 [\[\[CC1\]\]](#).
- **CC Part 2 Conformance:** This PP-Module is CC Part 2 extended, as it includes extended SFR components defined in [Extended Component Definitions](#).
- **CC Part 3 Conformance:** This PP-Module is CC Part 3 conformant, as it does not define extended SAR components.

5.2. Package Conformance

This PP-Module does not claim conformance to any pre-defined functional or assurance packages from CC:2022 Part 5 [\[\[CC5\]\]](#).

5.3. PP-Module Conformance Type

This PP-Module requires **Exact Conformance** from Security Targets claiming conformance.

5.3.1. Definition of Exact Conformance

Exact Conformance is a subset of Strict Conformance as defined by CC:2022. A Security Target demonstrating Exact Conformance to this PP-Module must satisfy the following requirements:

1. **Mandatory SFRs:** The ST shall include all SFRs specified in [Security Functional Requirements \(Mandatory\)](#) of this PP-Module.
2. **Selection-Based SFRs:** The ST may include SFRs from [Security Functional Requirements \(Selection-Based\)](#) when selections in mandatory SFRs trigger their inclusion, or when the TOE implements the corresponding functionality.
3. **Optional SFRs:** The ST may include SFRs from [Optional Requirements](#) as applicable to the TOE. (Note: This PP-Module currently defines no optional SFRs.)
4. **No Additional SFRs:** The ST shall not include any SFRs from CC Part 2 [\[\[CC2\]\]](#), CC Part 3 [\[\[CC3\]\]](#), or extended component definitions not already included in this PP-Module or the Base PP (cPP_DBMS).
5. **No Omissions:** No mandatory SFRs from [Security Functional Requirements \(Mandatory\)](#) may be omitted from the ST.
6. **Iteration Permitted:** Iteration of SFRs is permitted to address multiple instances of a security requirement when necessary.

7. **Operations:** All assignments, selections, and refinements shall be completed in accordance with the SFR specifications in this PP-Module and the Base PP.

5.4. Conformance Claim Rationale

A Security Target or PP claiming conformance to this PP-Module shall include:

1. A statement identifying this PP-Module by title, version, and publication date.
2. A statement that the claimed PP-Configuration includes the required Base PP (cPP_DBMS).
3. A statement that the claimed PP-Configuration includes the **DBMS Cryptographic Module**.
4. A statement that the ST demonstrates Exact Conformance to this PP-Module.

Chapter 6. Security Problem Definition

This section defines the security environment applicable to the TOE when deployed in a cloud environment. It is aligned with the CC in the Cloud Technical Community's model of a Trusted Platform and elastic, multitenant deployments.

The Security Problem Definition identifies threats that the TOE must counter, assumptions about the operational environment that must hold true for the TOE to operate securely, and organizational security policies that the TOE must enforce.

6.1. Threats

This section describes the threats that the TOE is designed to address. Each threat identifies the threat agent, the adverse action, and the asset being protected, in accordance with CC:2022 requirements.

T.ELASTIC_ENVIRONMENT

A malicious actor or an inadvertent system failure may exploit the dynamic, virtualized nature of cloud environments to cause the TOE to behave unpredictably or enter an insecure state. This may occur through manipulation of deployment topology changes, exploitation of ephemeral instance lifecycle events, or interference with scaling operations. The assets at risk include the TOE's security configuration, runtime state, and the confidentiality and integrity of protected data.

T.MALICIOUS_TENANT_ADMIN

An authorized administrator or orchestrator may intentionally or accidentally misconfigure the TOE, weaken security controls, or expose sensitive data. The TOE must enforce appropriate access control boundaries even when deployed by the tenant.

T.UNAUTHORIZED_ACCESS

An unauthorized user, including one with access to the underlying cloud infrastructure or control plane, may attempt to access TOE functionality or data without proper authentication or authorization.

T.INTEGRITY_FAILURE

A malicious actor may attempt to compromise the integrity of the TOE by launching or rehydrating it using tampered images, misconfigured templates, or unauthorized runtime components. In a cloud context, the threat agent may exploit weaknesses in image distribution, container registries, or deployment pipelines to substitute or modify TOE artifacts. The assets at risk include the TOE executable code, configuration data, and the trustworthiness of the deployed instance.

T.AUDIT_FAILURE

A malicious actor may attempt to interfere with the TOE's audit mechanisms to conceal unauthorized activities or security violations. The threat agent may exploit failures in audit generation, local buffering, or export mechanisms, particularly taking advantage of the ephemeral nature of cloud instances. Alternatively, system failures in elastic cloud

environments may inadvertently cause loss of audit data. The assets at risk include audit records necessary for security monitoring, incident investigation, and regulatory compliance.

T.SECRETS_EXPOSURE

A malicious actor may attempt to obtain cloud secrets, credentials, API keys, or other sensitive authentication material used by or stored within the TOE. The threat agent may exploit insecure secret storage, improper handling of environment variables, memory disclosure vulnerabilities, or unauthorized access to cloud secret management services. The assets at risk include database credentials, service account keys, encryption keys, and any other secrets that could enable unauthorized access to the TOE or its data.

T.NETWORK_INTERCEPTION

A malicious actor may attempt to intercept, modify, or inject data into communications between distributed TOE components or between the TOE and external services. In cloud environments, the threat agent may exploit shared network infrastructure, misconfigured network policies, or compromised adjacent workloads to perform network-based attacks. The assets at risk include TSF data transmitted between TOE nodes, management plane communications, and data exchanged with cloud-native services.

6.2. Assumptions

This section describes the assumptions about the operational environment that must be satisfied for the TOE to provide its security functionality. If these assumptions are not met, the TOE cannot be expected to operate securely.

A.TRUSTED_PLATFORM

The underlying cloud environment—including compute, network, storage, and orchestration infrastructure—is assumed to be a Trusted Platform as defined by the CCiTC model. Specifically, the platform must:

- Enforce isolation between tenants such that one tenant cannot access another tenant's compute resources, memory, storage, or network traffic;
- Protect the integrity of virtual machine or container boundaries against escape or breakout attacks;
- Provide accurate and unmodifiable identity assertions and policy information to the TOE;
- Implement resource allocation controls that prevent denial-of-service by co-located workloads.

This assumption aligns with the CCiTC Trusted Platform model and establishes the minimum isolation requirements for secure TOE operation.

A.SECURE_DEPLOYMENT

The TOE is deployed using an approved, authentic, and correctly configured image or installation method. Any orchestration system used must respect the security policies and runtime parameters defined for the TOE. Specifically:

- The deployment process must verify the cryptographic signature or hash of TOE artifacts

before installation;

- Configuration management tools must validate configuration parameters against a defined security baseline;
- Deployment logs must be generated and retained to support post-deployment verification;
- The deployment pipeline must be protected against unauthorized modification.

A.CLOUD_SERVICE_INTEGRITY

The cloud-native services that the TOE integrates with—such as IAM, key vaults, logging services, and configuration management systems—are assumed to operate correctly and provide accurate, untampered data to the TOE under normal operating conditions. In the event of partial service failures:

- The TOE will detect service unavailability through timeout or error responses;
- The TOE will fail securely by denying operations that depend on unavailable services rather than falling back to insecure defaults;
- The cloud provider will restore service availability within documented service level agreements.

This assumption does not extend to scenarios where cloud services are actively compromised by sophisticated threat actors.

A.AUTHORIZED_ADMINISTRATORS

Personnel with administrative access to the TOE, including tenant administrators and cloud audit authorities, are assumed to be:

- Appropriately vetted according to organizational security policies;
- Trained in the secure operation of the TOE and its cloud deployment environment;
- Bound by organizational policies that define acceptable use and prohibited actions;
- Subject to accountability mechanisms including activity logging and periodic access reviews.

This assumption does not preclude the possibility of insider threats but establishes that basic personnel security controls are in place.

A.TIME_SYNCHRONIZATION

The cloud platform provides reliable time synchronization services to the TOE, ensuring that:

- All TOE instances receive accurate time from a trusted source (e.g., cloud provider NTP service);
- Time drift between TOE instances and cloud services remains within acceptable bounds for cryptographic operations and audit timestamp accuracy;
- The TOE can rely on consistent timestamps for audit record generation, certificate validation, and time-based access controls.

6.3. Organizational Security Policies

This section defines the organizational security policies that the TOE must support or enforce. These policies represent security rules that organizations deploying the TOE are expected to implement.

P.AUDIT_RETENTION

The organization shall define and enforce audit record retention policies that specify:

- The minimum retention period for audit records based on regulatory requirements (e.g., GDPR, HIPAA, PCI-DSS, SOX) and organizational security policies;
- The storage location and protection requirements for retained audit records;
- Procedures for secure disposal of audit records after the retention period expires;
- Requirements for audit record availability during incident investigations or compliance audits.

The TOE must support the export of audit records to enable organizations to implement their retention policies using cloud-native or external audit storage solutions.

P.ACCESS_CONTROL_POLICY

The organization shall implement role-based access control with least-privilege principles for all TOE access. Specifically:

- Each user and service account shall be assigned to defined roles with explicitly documented privileges;
- Privileges shall be limited to the minimum necessary for users to perform their authorized functions;
- Administrative privileges shall be separated from regular user privileges;
- Access control decisions shall be logged for accountability;
- Role assignments shall be reviewed periodically and adjusted as organizational needs change.

The TOE must provide mechanisms to define roles, assign users to roles, and enforce access control decisions based on role membership.

Chapter 7. Security Objectives

This section defines the security objectives that address the threats, assumptions, and organizational security policies identified in the Security Problem Definition. Security objectives for the TOE describe what the TOE must do to counter threats and enforce policies. Security objectives for the Operational Environment describe what the environment must provide to support the TOE's security functionality.

7.1. Security Objectives for the TOE

O.CLOUD_IDENTITY_INTEGRATION

The TOE shall integrate with external cloud-native identity and access management (IAM) systems to authenticate users and map identities to internal roles. The TOE shall verify identity assertions from external providers and bind them to appropriate access control decisions.

O.CLOUD_AUDIT_INTEGRATION

The TOE shall generate audit records for security-relevant events and support export to cloud-native or tenant-managed audit collection systems to ensure observability and accountability. Audit records shall capture sufficient detail to support forensic analysis and compliance verification.

O.INTERNAL_RESOURCE_RESILIENCE

The TOE shall detect and signal disruptions affecting its distributed components, protect internal communications from disclosure and modification, and continue secure operation or fail securely when deployed in ephemeral, scaled, or orchestrated cloud environments. The TOE shall not claim to maintain full availability but shall ensure that security properties are preserved or that failure is handled in a manner that does not compromise the security posture.

Application Note: This objective focuses on detection, signaling, and fail-secure behavior rather than claiming to maintain confidentiality, integrity, and availability in all circumstances. The TOE provides awareness of disruptions and protects its internal data transfers, but availability guarantees depend on the operational environment (OE.TRUSTED_PLATFORM).

O.CLOUD_DEPLOYMENT_INTEGRITY

The TOE shall verify the integrity of its own runtime state or deployment artifacts (e.g., images, configurations) at startup or during operation, when such verification capabilities are implemented. The TOE shall reject tampered or unverifiable artifacts.

O.CONFIGURATION_PROTECTION

The TOE shall protect security-critical configuration from unauthorized or improper modification. The TOE shall enforce role-based restrictions on administrative actions affecting security functions and TSF data, preventing tenant administrators from weakening security controls beyond authorized limits.

Application Note: This objective directly addresses the threat of malicious or negligent tenant administrators (T.MALICIOUS_TENANT_ADMIN) who may attempt to weaken security configurations. The TOE must distinguish between legitimate administrative actions and those

that would compromise security, enforcing appropriate restrictions based on assigned roles.

7.2. Security Objectives for the Operational Environment

OE.TRUSTED_PLATFORM

The platform on which the TOE is deployed (including the hypervisor, control plane, and IAM services) shall enforce tenant isolation, protect compute and storage resources, and provide trusted identity and configuration assertions. The Trusted Platform is responsible for availability of underlying resources and correct operation of infrastructure services.

OE.SECURE_ORCHESTRATION

Any orchestration toolchain or deployment service used to provision the TOE (e.g., Terraform, Helm, cloud-init) shall use authentic and approved TOE artifacts and preserve security-relevant configuration parameters. Orchestration systems shall not introduce unauthorized modifications to the TOE configuration.

OE.LOGGING_SERVICES

If audit export is selected (FAU_STG_EXT.1), the operational environment shall provide a trusted logging pipeline or SIEM that reliably receives and stores exported audit records. The logging service shall protect audit data from unauthorized modification or deletion.

Conditional Trigger: This objective applies when the ST author selects FAU_STG_EXT.1 (Audit Export) to export audit data to external cloud-native logging or SIEM services.

OE.IDENTITY_PROVIDERS

If federated authentication is used (as selected in FIA_UID_EXT.1), the operational environment shall provide an identity provider that issues correct, verifiable assertions consumed by the TOE. The identity provider shall authenticate users according to organizational policy and provide accurate role and attribute information.

Conditional Trigger: This objective applies when the ST author selects "federated directory users" or "cloud IAM roles" in FIA_UID_EXT.1 (External Identity Integration).

OE.SECRETS_MANAGEMENT

The operational environment shall ensure that secrets provided to the TOE are authentic and protected from unauthorized disclosure. This includes API tokens, database connection strings, and cryptographic keys used for decrypting configuration data.

If the TOE claims to import cryptographic keys (as selected in **FCS_CKM_EXT.1** of the DBMS Crypto Module), the secret management service shall securely provide those keys to the TOE via a trusted channel defined by **FDP_ITC_EXT.1**.

Conditional Trigger: This objective applies when the TOE integrates with cloud-native secret management systems or when **FCS_CKM_EXT.1** (Imported Keys) is selected in the DBMS Crypto Module.

Application Note: This objective distinguishes between "configuration secrets" (covered here)

and "Data-at-Rest Cryptographic Keys". While configuration secrets are managed here, the lifecycle of Data-at-Rest Keys is managed by the DBMS Crypto Module. However, if the Master Key is stored in a Cloud Secret Manager (e.g., AWS Secrets Manager), this objective ensures the platform protects that secret store.

Chapter 8. Security Rationale

This section provides rationale showing how the defined security objectives address the identified threats and assumptions, and how the Security Functional Requirements (SFRs) in this module satisfy the objectives. The rationale demonstrates completeness and consistency of the security problem definition, objectives, and requirements.

8.1. Threats to Objectives Mapping

The following table maps each threat to the security objectives that address it. Each threat must be fully addressed by the combination of TOE objectives and operational environment objectives.

Threat	Security Objectives Addressing the Threat	Rationale
T.ELASTIC_ENVIRONMENT	O.INTERNAL_RESOURCE_RESILIENCE OE.TRUSTED_PLATFORM OE.SECURE_ORCHESTRATION	The TOE detects and signals disruptions, protects internal communications, and fails securely (O.INTERNAL_RESOURCE_RESILIENCE). The platform ensures resource protection and isolation (OE.TRUSTED_PLATFORM). Orchestration preserves security parameters across deployments (OE.SECURE_ORCHESTRATION).
T.MALICIOUS_TENANT_ADMIN	O.CLOUD_IDENTITY_INTEGRATION O.CLOUD_AUDIT_INTEGRATION O.CONFIGURATION_PROTECTION OE.SECURE_ORCHESTRATION	The TOE enforces proper authentication and role mapping (O.CLOUD_IDENTITY_INTEGRATION), records administrative actions for accountability (O.CLOUD_AUDIT_INTEGRATION), and restricts configuration changes to authorized roles (O.CONFIGURATION_PROTECTION). Orchestration systems respect security policies (OE.SECURE_ORCHESTRATION).
T.UNAUTHORIZED_ACCESS	O.CLOUD_IDENTITY_INTEGRATION OE.TRUSTED_PLATFORM OE.IDENTITY_PROVIDERS	The TOE verifies user authentication via cloud IAM (O.CLOUD_IDENTITY_INTEGRATION). The platform enforces tenant isolation (OE.TRUSTED_PLATFORM). Identity providers supply accurate assertions (OE.IDENTITY_PROVIDERS).
T.INTEGRITY_FAILURE	O.CLOUD_DEPLOYMENT_INTEGRITY OE.SECURE_ORCHESTRATION OE.TRUSTED_PLATFORM	The TOE verifies runtime state and deployment artifacts (O.CLOUD_DEPLOYMENT_INTEGRITY). Orchestration uses authentic and approved artifacts (OE.SECURE_ORCHESTRATION). The platform protects compute resources (OE.TRUSTED_PLATFORM).

Threat	Security Objectives Addressing the Threat	Rationale
T.AUDIT_FAIL URE	O.CLOUD_AUDIT_INTEGRATION OE.LOGGING_SERVICES	The TOE generates audit records and supports export (O.CLOUD_AUDIT_INTEGRATION). The OE provides reliable audit storage (OE.LOGGING_SERVICES).
T.SECRETS_EXP OSURE	O.CONFIGURATION_PROTECTI ON OE.SECRETS_MANAGEMENT O.CLOUD_DEPLOYMENT_INTEG RITY Crypto Module: FCS_CKM_EXT.1	The TOE protects security-critical configuration including credential handling (O.CONFIGURATION_PROTECTION). The OE ensures secrets are authentic and protected (OE.SECRETS_MANAGEMENT). If keys are imported, the Crypto Module ensures they are received via a trusted channel (FCS_CKM_EXT.1 referencing FDP_ITC_EXT.1).
T.NETWORK_I NTERCEPTION	O.INTERNAL_RESOURCE_RESIL IENCE OE.TRUSTED_PLATFORM Crypto Module: FDP_DIT_EXT.1	The TOE protects internal communications from disclosure and modification (O.INTERNAL_RESOURCE_RESILIENCE). The platform provides network isolation (OE.TRUSTED_PLATFORM). External network protection (e.g., Client-to-DB, DB-to-KMS) is addressed by FDP_DIT_EXT.1 (TLS) defined in the DBMS Crypto Module.

8.2. Assumptions to Objectives Mapping

The following table maps each assumption to the operational environment objectives that uphold it.

Assumption	OE Objectives Upholding the Assumption	Rationale
A.TRUSTED_PL ATFORM	OE.TRUSTED_PLATFORM	The operational environment objective OE.TRUSTED_PLATFORM directly requires the platform to enforce tenant isolation, protect resources, and provide trusted identity assertions, which are the exact properties assumed in A.TRUSTED_PLATFORM.
A.SECURE_DEP LOYMENT	OE.SECURE_ORCHESTRATION	The operational environment objective OE.SECURE_ORCHESTRATION requires orchestration systems to use authentic artifacts and preserve security parameters, directly upholding the assumption that deployment is secure.

Assumption	OE Objectives Upholding the Assumption	Rationale
A.CLOUD_SERVICE_INTEGRITY	OE.IDENTITY_PROVIDERS OE.LOGGING_SERVICES OE.SECRETS_MANAGEMENT	The assumption that cloud services operate correctly is upheld by: OE.IDENTITY_PROVIDERS (correct identity assertions), OE.LOGGING_SERVICES (reliable audit services), and OE.SECRETS_MANAGEMENT (authentic secrets protected from disclosure).

8.3. Objectives to SFRs Mapping

The following table maps each TOE security objective to the SFRs that satisfy it. Each objective must be fully satisfied by the combination of mandatory and selection-based SFRs.

Security Objective for the TOE	SFRs Satisfying the Objective	Rationale
O.CLOUD_IDENTITY_INTEGRATION	FIA_UID_EXT.1 FIA_UAU.1 FMT_SMR.1/Cloud FMT_SMF.1/Cloud FMT_MTD_EXT.1 Crypto Module: FDP_DIT_EXT.1	FIA_UID_EXT.1 enables external identity integration. FDP_DIT_EXT.1 (from the Crypto Module) ensures the connection to the Identity Provider is cryptographically protected.
O.CLOUD_AUDIT_INTEGRATION	FAU_GEN.1 FAU_SEL.1/Cloud FAU_STG.1 FAU_STG_EXT.1 FMT_SMF.1/Cloud Crypto Module: FDP_DIT_EXT.1	FAU_STG_EXT.1 enables export. FDP_DIT_EXT.1 (from the Crypto Module) ensures the export channel to the SIEM is cryptographically protected.
O.INTERNAL_RESOURCE_RESILIENCE	FPT_ARS_EXT.1 FPT_ITT.1 FPT_FLS.1 (inherited from Base PP) FPT_TST.1 (if selected)	FPT_ARS_EXT.1 monitors runtime environment and signals disruptions. FPT_ITT.1 protects internal TSF data transfers from disclosure and modification. FPT_FLS.1 (from Base PP) ensures fail-secure behavior. FPT_TST.1 validates integrity via self-test when selected. + Note: This objective focuses on detection, signaling, and secure failure rather than availability guarantees. The combination of these SFRs ensures the TOE can detect problems, protect communications, and fail securely.

Security Objective for the TOE	SFRs Satisfying the Objective	Rationale
O.CLOUD_DEPLOYMENT_INTEGRITY	FPT_TST.1 (if selected) FPT_SBT_EXT.1 (if selected) FPT_TUD_EXT.1	FPT_TST.1 validates runtime integrity via self-test. FPT_SBT_EXT.1 verifies boot-time and deployment artifacts. FPT_TUD_EXT.1 verifies authenticity and integrity of updates.
O.CONFIGURATION_PROTECTION	FMT_MOF.1 FMT_MTD_EXT.1 FMT_SMR.1/Cloud FMT_SMF.1/Cloud Crypto Module: FCS_CKM_EXT.1	FMT_MOF.1 restricts management of security functions. If the TOE uses imported keys for configuration encryption, FCS_CKM_EXT.1 (from the Crypto Module) ensures their secure retrieval.

8.4. OE Objectives Conditional Triggers Summary

The following table summarizes when each conditional OE objective applies based on ST selections.

OE Objective	Triggering Selection	Triggering SFR
OE.LOGGING_SERVICES	Audit export to cloud-native or SIEM services	FAU_STG_EXT.1 (Audit Export)
OE.IDENTITY_PROVIDERS	Federated directory users or cloud IAM roles for authentication	FIA_UID_EXT.1 (External Identity Integration) - selection of "cloud IAM roles" or "federated directory users"
OE.SECRETS_MANAGEMENT	Cloud-native secret management integration	FMT_MTD_EXT.1 (Cloud Configuration State Protection) - when assignment includes cloud secret management systems

8.5. Consistency Summary

8.5.1. Threat Coverage

All threats defined in the Security Problem Definition are addressed by at least one security objective:

- T.ELASTIC_ENVIRONMENT: Addressed by O.INTERNAL_RESOURCE_RESILIENCE, OE.TRUSTED_PLATFORM, OE.SECURE_ORCHESTRATION
- T.MALICIOUS_TENANT_ADMIN: Addressed by O.CLOUD_IDENTITY_INTEGRATION, O.CLOUD_AUDIT_INTEGRATION, OE.SECURE_ORCHESTRATION
- T.UNAUTHORIZED_ACCESS: Addressed by O.CLOUD_IDENTITY_INTEGRATION, OE.TRUSTED_PLATFORM, OE.IDENTITY_PROVIDERS
- T.INTEGRITY_FAILURE: Addressed by O.CLOUD_DEPLOYMENT_INTEGRITY,

OE.SECURE_ORCHESTRATION, OE.TRUSTED_PLATFORM

- T.AUDIT_FAILURE: Addressed by O.CLOUD_AUDIT_INTEGRATION, OE.LOGGING_SERVICES
- T.SECRETS_EXPOSURE: Addressed by O.CONFIGURATION_PROTECTION, OE.SECRETS_MANAGEMENT, OE.TRUSTED_PLATFORM
- T.NETWORK_INTERCEPTION: Addressed by O.INTERNAL_RESOURCE_RESILIENCE, OE.TRUSTED_PLATFORM (plus selection-based FDP_DIT_EXT.1)

8.5.2. Assumption Coverage

All assumptions are upheld by operational environment objectives:

- A.TRUSTED_PLATFORM: Upheld by OE.TRUSTED_PLATFORM
- A.SECURE_DEPLOYMENT: Upheld by OE.SECURE_ORCHESTRATION
- A.CLOUD_SERVICE_INTEGRITY: Upheld by OE.IDENTITY_PROVIDERS, OE.LOGGING_SERVICES, OE.SECRETS_MANAGEMENT

8.5.3. Objective Coverage

All TOE security objectives are satisfied by SFRs:

- O.CLOUD_IDENTITY_INTEGRATION: FIA_UID_EXT.1, FIA_UAU.1, FMT_SMR.1/Cloud, FMT_SMF.1/Cloud, FMT_MTD_EXT.1
- O.CLOUD_AUDIT_INTEGRATION: FAU_GEN.1, FAU_SEL.1/Cloud, FAU_STG.1, FAU_STG_EXT.1, FMT_SMF.1/Cloud
- O.INTERNAL_RESOURCE_RESILIENCE: FPT_ARS_EXT.1, FPT_ITT.1, FPT_FLS.1, FPT_TST.1
- O.CLOUD_DEPLOYMENT_INTEGRITY: FPT_TST.1, FPT_SBT_EXT.1, FPT_TUD_EXT.1
- O.CONFIGURATION_PROTECTION: FMT_MOF.1, FMT_MTD_EXT.1, FMT_SMR.1/Cloud, FMT_SMF.1/Cloud

All operational environment objectives are either unconditional or have clear conditional triggers linked to specific SFR selections.

8.5.4. Cryptographic Deferral

This PP-Module does not define cryptographic objectives or SFRs. All cryptographic protections (data-in-transit via TLS/HTTPS, data-at-rest encryption) are addressed through selection-based SFRs (FDP_DIT_EXT.1, FDP_DAR_EXT.1) that invoke the DBMS Crypto Module (DBMS_MOD_CRYPT0). This separation maintains clear boundaries between:

- This module: Cloud integration, resilience, configuration protection
- DBMS_MOD_CRYPT0: Cryptographic algorithms, key management, protocol implementation

Chapter 9. Security Functional Requirements

9.1. Conventions

The individual security functional requirements are specified in the sections below. The following conventions are used for the completion of operations:

- [*Italicized text within square brackets*] indicates an operation to be completed by the ST author.
- **Bold text** indicates additional text provided as a refinement.
- [**Bold text within square brackets**] indicates the completion of an assignment.
- [text within square brackets] indicates the completion of a selection.
- Number in parentheses after SFR name, e.g. (1) indicates the completion of an iteration.

Extended SFRs are identified by having a label "EXT" at the end of the SFR name.

Chapter 10. Security Functional Requirements (Mandatory)

The following SFRs are mandatory for all TOEs claiming conformance to this DBMS in the Cloud Module. They extend or refine SFRs defined in the DBMS Base PP, aligned with the cloud deployment context and the CCiTC Trusted Platform model.

10.1. FAU: Security Audit

10.1.1. FAU_GEN.1 Audit Data Generation

FAU_GEN.1.1 The TSF shall be able to generate an audit record of the following auditable events:

a) Start-up and shutdown of the audit functions; b) All auditable events for the [selection: minimum, basic, detailed, not specified] level of audit; and c) [assignment: *other specifically defined auditable events*].

Application Note: The ST author shall select the level of audit appropriate for the TOE and specify any additional cloud-specific auditable events. Cloud-specific auditable events should include, at minimum:

- Administrative actions via cloud interfaces
- API interactions with the TOE
- Configuration or deployment changes
- Detected failures or unavailability events
- External identity provider authentication events
- Changes to cloud secret management integrations

FAU_GEN.1.2 The TSF shall record within each audit record at least the following information:

a) Date and time of the event, type of event, subject identity (if applicable), and the outcome (success or failure) of the event; and b) For each audit event type, based on the auditable event definitions of the functional components included in the PP/ST, [assignment: *other audit relevant information*].

Application Note: Cloud deployments should include tenant identifier, cloud region/zone, and source IP address where applicable.

10.1.1.1. Dependencies

Dependency	Resolution
FPT_STM.1 Reliable time stamps	Satisfied by A.TIME_SYNCHRONIZATION. The operational environment (Trusted Platform) provides reliable time synchronization services. Cloud providers supply synchronized time sources (e.g., NTP, cloud-native time services) that the TOE consumes. This is appropriate because time synchronization is a fundamental cloud platform service outside the TOE boundary.

10.1.2. FAU_SEL.1/Cloud Selective Audit (Cloud)

FAU_SEL.1/Cloud.1 The TSF shall be able to select the set of events to be audited from the set of all auditable events based on the following attributes:

a) [selection: object identity, user identity, subject identity, host identity, event type]; b) [assignment: *list of additional attributes on which audit selectivity is based*].

Application Note: For cloud deployments, the ST author should include the following additional attributes for audit selectivity:

- Tenant identifier
- IAM role or cloud identity
- API endpoint or operation type
- Cloud region or availability zone
- Source network range

FAU_SEL.1/Cloud.2 The TSF shall provide the ability to [selection: include, exclude] auditable events from the set of audited events based on the defined selection criteria.

10.1.2.1. Dependencies

Dependency	Resolution
FAU_GEN.1 Audit data generation	Satisfied by FAU_GEN.1 in this PP-Module.
FMT_MTD.1 Management of TSF data	Satisfied by FMT_MTD_EXT.1 in this PP-Module which provides management of TSF data in cloud contexts.

10.1.3. FAU_STG.1 Protected Audit Trail Storage

FAU_STG.1.1 The TSF shall [selection: store audit records on the TOE, transmit audit records to an external IT entity using a trusted channel as specified in FTP_ITC.1, *[assignment: other storage location]*].

Application Note: Cloud deployments may store audit records locally prior to export, or transmit directly to external audit collection systems. When transmitting to external systems, a trusted channel should be used.

FAU_STG.1.2 The TSF shall protect the stored audit records in the audit trail from unauthorized deletion.

FAU_STG.1.3 The TSF shall be able to [selection: prevent, detect] unauthorized modifications to the stored audit records in the audit trail.

Application Note: This SFR applies to local audit storage prior to export or cloud integration. When FAU_STG_EXT.1 (Audit Export) is selected, the operational environment (OE.LOGGING_SERVICES) is responsible for protecting exported audit records.

10.1.3.1. Dependencies

Dependency	Resolution
FAU_GEN.1 Audit data generation	Satisfied by FAU_GEN.1 in this PP-Module.

10.1.4. FAU Class Dependencies Summary

Table 3. FAU Class Dependency Resolution

SFR	Dependencies	Resolution
FAU_GEN.1	FPT_STM.1	A.TIME_SYNCHRONIZATION (operational environment)
FAU_SEL.1/Cloud	FAU_GEN.1, FMT_MTD.1	FAU_GEN.1 (this module), FMT_MTD_EXT.1 (this module)
FAU_STG.1	FAU_GEN.1	FAU_GEN.1 (this module)

10.2. FIA: Identification and Authentication

10.2.1. FIA_UAU.1 Timing of Authentication

FIA_UAU.1.1 The TSF shall allow [assignment: *list of TSF-mediated actions*] on behalf of the user to be performed before the user is authenticated.

Application Note: Actions typically allowed before authentication include:

- Viewing public connection endpoints or service health status
- Initiating the authentication handshake itself
- Accessing unauthenticated documentation or help resources

The ST author shall enumerate all actions permitted before authentication. This list should be minimal and not include access to any protected data or administrative functions.

FIA_UAU.1.2 The TSF shall require each user to be successfully authenticated before allowing any other TSF-mediated actions on behalf of that user.

Application Note: Authentication itself may be delegated to external cloud IAM systems; however,

the TOE must verify and enforce the authentication result before permitting access. When external identity providers are used (as specified in FIA_UID_EXT.1), the TOE must validate the authentication assertions received.

10.2.1.1. Dependencies

Dependency	Resolution
FIA_UID.1 Timing of identification	Satisfied by FIA_UID_EXT.1 in this PP-Module. FIA_UID_EXT.1 extends FIA_UID.1 to support cloud-native identity providers and provides equivalent or greater identification capability.

10.2.2. FIA_UID_EXT.1 External Identity Integration

FIA_UID_EXT.1.1 The TSF shall allow [assignment: *list of TSF-mediated actions*] on behalf of the user to be performed before the user is identified.

Application Note: This should be consistent with actions allowed before authentication in FIA_UAU.1.1. Communication between the TOE and the External Identity Provider shall be protected by a trusted channel. The ST author shall select TLS in FDP_DIT_EXT.1 of the DBMS Crypto Module to satisfy this protection requirement.

FIA_UID_EXT.1.2 The TSF shall require each user to be successfully identified before allowing any other TSF-mediated actions on behalf of that user.

FIA_UID_EXT.1.3 The TSF shall provide user identification using external identity providers such as [selection: cloud IAM roles, federated directory users (e.g., LDAP/AD integrated with cloud IAM), service principals, [assignment: *other identity sources*]].

Application Note: This requirement ensures the TOE can interpret identities from external cloud-native IAM systems and map them to internal user representations. The TOE must validate identity assertions from external providers before granting access.

10.2.2.1. Dependencies

Dependency	Resolution
FIA_UAU.1 Timing of Authentication	Satisfied by FIA_UAU.1 in this PP-Module.

10.2.3. FIA Class Dependencies Summary

Table 4. FIA Class Dependency Resolution

SFR	Dependencies	Resolution
FIA_UAU.1	FIA_UID.1	FIA_UID_EXT.1 (this module, hierarchical to FIA_UID.1)
FIA_UID_EXT.1	FIA_UAU.1	FIA_UAU.1 (this module)

10.3. FMT: Security Management

10.3.1. FMT_MOF.1 Management of Security Functions Behavior

FMT_MOF.1.1 The TSF shall restrict the ability to [selection: determine the behaviour of, disable, enable, modify the behaviour of] the functions [assignment: *list of functions*] to [assignment: *the authorized identified roles*].

Application Note: The ST author shall specify:

- The functions to be managed (should include those specified in FMT_SMF.1 and FMT_SMF.1/Cloud)
- The authorized roles permitted to manage these functions (should include tenant administrator and cloud audit authority as defined in FMT_SMR.1/Cloud)

This ensures cloud-based management functions remain under strict role-based access control.

10.3.1.1. Dependencies

Dependency	Resolution
FMT_SMF.1 Specification of Management Functions	Satisfied by FMT_SMF.1/Cloud in this PP-Module.
FMT_SMR.1 Security roles	Satisfied by FMT_SMR.1/Cloud in this PP-Module.

10.3.2. FMT_MTD_EXT.1 Cloud Configuration State Protection

FMT_MTD_EXT.1.1 The TSF shall restrict the ability to [selection: query, modify, delete, clear, [assignment: *other operations*]] the [assignment: *list of TSF data*] to [assignment: *the authorized identified roles*].

FMT_MTD_EXT.1.2 The TSF shall manage and protect TSF data stored in cloud-native configuration and secret management systems [assignment: *list of supported cloud secret management systems (e.g., AWS Secrets Manager, Azure Key Vault, Oracle Key Vault, HashiCorp Vault, environment variables)*].

Application Note: This requirement ensures sensitive data, such as database credentials, encryption keys, and configuration parameters, is handled securely when stored in external cloud-native systems. The TOE should support secure retrieval and protect credentials in memory after retrieval. Retrieval of configuration data or secrets from cloud secret management systems shall be protected. If the TOE performs this retrieval, the ST author shall claim FDP_DIT_EXT.1 (TLS) from the DBMS Crypto Module. If the retrieved data is a cryptographic key, the ST author shall also claim FCS_CKM_EXT.1 (Imported Keys) from the DBMS Crypto Module.

10.3.2.1. Dependencies

Dependency	Resolution
FMT_SMF.1 Specification of Management Functions	Satisfied by FMT_SMF.1/Cloud in this PP-Module.
FMT_SMR.1 Security roles	Satisfied by FMT_SMR.1/Cloud in this PP-Module.

10.3.3. FMT_SMF.1/Cloud Specification of Management Functions (Cloud)

FMT_SMF.1/Cloud.1 The TSF shall be capable of performing the following management functions:

a) Configuration of audit selection criteria as specified by FAU_SEL.1/Cloud; b) Configuration of audit export destinations as specified by FAU_STG_EXT.1 (when selected); c) Configuration of secure external communications (TLS/HTTPS) as specified by FDP_DIT_EXT.1 (when selected); d) Configuration of data-at-rest encryption as specified by FDP_DAR_EXT.1 (when selected); e) Management of external identity provider mappings as specified by FIA_UID_EXT.1; f) Monitoring and querying deployment integrity status; g) Rotation of cloud-managed credentials or secrets; h) [assignment: *other cloud-specific management functions*].

Application Note: This SFR explicitly invokes selection-based SFRs FAU_STG_EXT.1, FDP_DIT_EXT.1, and FDP_DAR_EXT.1 when the TOE implements these capabilities. The ST author should ensure management functions are available for all claimed security functions.

10.3.3.1. Dependencies

Dependency	Resolution
No dependencies	N/A

10.3.4. FMT_SMR.1/Cloud Security Roles (Cloud)

FMT_SMR.1/Cloud.1 The TSF shall maintain the roles [assignment: *the authorized identified roles (e.g., tenant administrator, tenant user, cloud audit authority)*].

FMT_SMR.1/Cloud.2 The TSF shall be able to associate users with roles.

Application Note: This requirement extends base roles for cloud-specific use cases such as tenant-scoped administration. Roles should be mapped from external identity providers where applicable. The cloud audit authority role provides read-only access to audit data and security status for compliance purposes.

10.3.4.1. Dependencies

Dependency	Resolution
FIA_UID.1 Timing of identification	Satisfied by FIA_UID_EXT.1 in this PP-Module.

10.3.5. FMT Class Dependencies Summary

Table 5. FMT Class Dependency Resolution

SFR	Dependencies	Resolution
FMT_MOF.1	FMT_SMF.1, FMT_SMR.1	FMT_SMF.1/Cloud (this module), FMT_SMR.1/Cloud (this module)
FMT_MTD_EXT.1	FMT_SMF.1, FMT_SMR.1	FMT_SMF.1/Cloud (this module), FMT_SMR.1/Cloud (this module)
FMT_SMF.1/Cloud	None	N/A
FMT_SMR.1/Cloud	FIA_UID.1	FIA_UID_EXT.1 (this module)

10.4. FPT: Protection of the TSF

10.4.1. FPT_ITT.1 Basic Internal TSF Data Transfer Protection

FPT_ITT.1.1 The TSF shall protect TSF data from [selection: disclosure, modification, disclosure and modification] when it is transmitted between separate parts of the TOE.

Application Note: This requirement applies explicitly to distributed deployments, clusters, or management/data plane separations within cloud deployments. Examples include:

- Communication between database nodes in a cluster
- Communication between control plane and data plane components
- Replication traffic between geographically distributed instances

When "disclosure" is selected, confidentiality protection (encryption) is required. When "modification" is selected, integrity protection is required. The specific cryptographic mechanisms should be defined in the DBMS Crypto Module when cryptographic protection is used.

10.4.1.1. Dependencies

Dependency	Resolution
No dependencies specified in CC	The PP-Module recommends but does not require cryptographic mechanisms defined in the DBMS Crypto Module for implementations selecting protection from disclosure.

10.4.2. FPT_ARS_EXT.1 Availability and Resilience Signaling

FPT_ARS_EXT.1.1 The TSF shall monitor its runtime environment for indicators of [selection: configuration drift, resource unavailability, integrity failure via self-test, orchestration failure, [assignment: *other monitored conditions*]].

FPT_ARS_EXT.1.2 The TSF shall notify [assignment: *authorized administrator, orchestrator, or monitoring system*] upon detection of a monitored condition.

Application Note: If "integrity failure via self-test" is selected, the TOE must implement FPT_TST.1 accordingly. This requirement provides awareness and alerting capability in cloud deployments where infrastructure state may change dynamically.

10.4.2.1. Dependencies

Dependency	Resolution
No dependencies	N/A
Conditional: FPT_TST.1 (when "integrity failure via self-test" is selected)	See Selection-Based Requirements

10.4.3. FPT_TUD_EXT.1 Trusted Update

FPT_TUD_EXT.1.1 The TSF shall provide authorized users the ability to query the current version of the TOE software/firmware.

FPT_TUD_EXT.1.2 The TSF shall provide a mechanism to apply updates using [assignment: *cloud-native deployment services, trusted image registries, secure boot mechanisms as defined in FPT_SBT_EXT.1*].

FPT_TUD_EXT.1.3 The TSF shall verify the authenticity and integrity of updates prior to installation using [assignment: *digital signature verification, cryptographic hash verification, other mechanisms*].

Application Note: Selecting "secure boot mechanisms as defined in FPT_SBT_EXT.1" explicitly triggers the requirement to claim FPT_SBT_EXT.1.

10.4.3.1. Dependencies

Dependency	Resolution
Conditional: FPT_SBT_EXT.1 (when secure boot is selected)	See Selection-Based Requirements

10.4.4. FPT Class Dependencies Summary

Table 6. FPT Class Dependency Resolution

SFR	Dependencies	Resolution
FPT_ITT.1	None (CC)	N/A
FPT_ARS_EXT.1	None; conditional on FPT_TST.1	FPT_TST.1 (selection-based, when self-test selected)
FPT_TUD_EXT.1	Conditional on FPT_SBT_EXT.1	FPT_SBT_EXT.1 (selection-based, when secure boot selected)

10.5. Time Stamps Dependency Rationale

10.5.1. A.TIME_SYNCHRONIZATION (Assumption for FPT_STM.1 Dependency)

The FAU_GEN.1 requirement depends on FPT_STM.1 (Reliable time stamps) to provide accurate

timestamps for audit records. This PP-Module satisfies this dependency through the operational environment assumption A.TIME_SYNCHRONIZATION, rather than including FPT_STM.1 as a mandatory SFR.

Rationale:

1. **Cloud Platform Responsibility:** In cloud deployment contexts, time synchronization is a fundamental infrastructure service provided by the Trusted Platform. Cloud providers supply highly accurate, synchronized time sources (e.g., AWS Time Sync Service, Azure Time Service, Google Cloud NTP) that are available to all compute instances.
2. **TOE Boundary:** Including time synchronization functionality within the TOE boundary would be redundant with cloud platform capabilities and would unnecessarily expand the evaluation scope.
3. **Trusted Platform Model:** Per the CCiTC architectural model, the Trusted Platform provides reliable infrastructure services including time synchronization. The TOE is entitled to rely on these services.
4. **Operational Environment Control:** The operational environment requirements (OE.TRUSTED_PLATFORM) already require the platform to provide trusted services to the TOE, which includes accurate time.

Assumption Statement:

A.TIME_SYNCHRONIZATION

The operational environment provides reliable, synchronized time services to the TOE. Time sources provided by the Trusted Platform are accurate and protected from unauthorized modification. The TOE consumes time services from the operational environment for generating timestamps in audit records.

Application Note: The ST author should ensure guidance documentation instructs administrators to verify that the cloud platform's time synchronization services are properly configured and that the TOE is configured to use these services.

Chapter 11. Security Assurance Requirements (SARs)

This PP-Module does not define additional Security Assurance Requirements beyond those already specified by the collaborative Protection Profile for Database Management Systems (cPP_DBMS).

All SARs defined in the cPP_DBMS apply directly and fully to TOEs claiming conformance with this DBMS Cloud Module.

11.1. SAR Inheritance

The following SAR families, as defined in the cPP_DBMS, are inherited unchanged:

- ADV: Development
- AGD: Guidance Documents
- ALC: Life-cycle Support
- ASE: Security Target Evaluation
- ATE: Tests
- AVA: Vulnerability Assessment

No additional or modified SARs are introduced by this PP-Module.

Appendix A: Selection-Based Requirements

Chapter 12. Security Functional Requirements (Selection-Based)

These SFRs apply only when the TOE implements the specified functionality or when triggered by selections in mandatory SFRs. When selected, these SFRs may invoke additional requirements or modules (e.g., DBMS Crypto Module).

12.1. FAU: Security Audit

12.1.1. FAU_STG_EXT.1 Audit Export

FAU_STG_EXT.1.1 The TSF shall support export of audit data to [assignment: *cloud-native logging or SIEM services (e.g., AWS CloudWatch, Azure Monitor, Splunk, tenant-owned SIEM solutions)*].

FAU_STG_EXT.1.2 The TSF shall export audit data using [selection: a trusted channel as specified in FTP_ITC.1, [assignment: *other secure export mechanism*]].

Application Note: This SFR is selected when the management function "Configuration of audit export destinations" is claimed in FMT_SMF.1/Cloud. Selection implies reliance on external audit storage as specified in OE.LOGGING_SERVICES.

12.1.1.1. Dependencies

Dependency	Resolution
FAU_GEN.1 Audit data generation	Satisfied by FAU_GEN.1 (mandatory, this module).
FAU_STG.1 Protected audit trail storage	Satisfied by FAU_STG.1 (mandatory, this module).

12.2. FDP: User Data Protection

12.2.1. FDP_DAR_EXT.1 Data-at-Rest Protection

FDP_DAR_EXT.1.1 The TSF shall protect persistent storage of [assignment: *sensitive data types*] using [selection: encryption as defined in FCS_COP.1 and FCS_CKM.1 of the DBMS Crypto Module, disk encryption managed externally by the operational environment, none].

Application Note: Selecting "encryption as defined in FCS_COP.1 and FCS_CKM.1 of the DBMS Crypto Module" invokes a requirement to include the DBMS Crypto Module in the PP-Configuration. The option "none" must be justified by the lack of sensitive data persistence or by external security mitigations documented in the ST.

12.2.1.1. Dependencies

Dependency	Resolution
Conditional: FCS_COP.1, FCS_CKM.1 (when TOE-implemented encryption selected)	Satisfied by DBMS Crypto Module (must be included in PP-Configuration).
None (when external disk encryption or none selected)	N/A

12.2.2. FDP_DIT_EXT.1 Data-in-Transit Protection

FDP_DIT_EXT.1.1 The TSF shall protect data in transit between the TOE and external entities using [selection: TLS as defined in FCS_TLSC_EXT.1 and FCS_TLSS_EXT.1 of the DBMS Crypto Module, HTTPS as defined in FCS_HTTPS_EXT.1 of the DBMS Crypto Module, [assignment: *other cryptographic protocols*]].

Application Note: Selecting TLS or HTTPS invokes a requirement to include the DBMS Crypto Module via the PP-Configuration. This SFR applies to management APIs, admin consoles, and remote DBMS connections exposed to untrusted networks.

Application Note: Because this module is claimed in conjunction with the DBMS Cloud Module, the selection of "TLS" or "HTTPS" in this SFR is effectively mandatory to satisfy the cloud integration requirements (e.g., protecting the channel to the Cloud IAM provider).

12.2.2.1. Dependencies

Dependency	Resolution
FCS_TLSC_EXT.1, FCS_TLSS_EXT.1 (when TLS selected)	Satisfied by DBMS Crypto Module (must be included in PP-Configuration).
FCS_HTTPS_EXT.1 (when HTTPS selected)	Satisfied by DBMS Crypto Module (must be included in PP-Configuration).

12.3. FPT: Protection of the TSF

12.3.1. FPT_SBT_EXT.1 Secure Boot and Image Verification

FPT_SBT_EXT.1.1 The TSF shall verify the integrity of its boot-time or deployment-time components using [assignment: *verification mechanisms (e.g., digital signature verification, cryptographic hash verification)*].

FPT_SBT_EXT.1.2 The TSF shall verify the authenticity of its boot-time or deployment-time components by validating [selection: digital signatures from trusted signing authorities, cryptographic hashes against known-good values, [assignment: *other authenticity verification methods*]].

FPT_SBT_EXT.1.3 The TSF shall [selection: prevent boot/startup, alert administrators, [assignment:

other action]] if integrity or authenticity verification fails.

Application Note: This SFR applies primarily to containerized deployments, appliance images, or orchestrated workloads with explicit image trust requirements. It is triggered when "secure boot mechanisms as defined in FPT_SBT_EXT.1" is selected in FPT_TUD_EXT.1.

12.3.1.1. Dependencies

Dependency	Resolution
No dependencies	N/A

12.3.2. FPT_TST.1 TSF Self-Test

FPT_TST.1.1 The TSF shall run a suite of self tests [selection: during initial start-up, periodically during normal operation, at the request of the authorized user, at the conditions [assignment: *conditions under which self test should occur*]] to demonstrate the correct operation of [selection: [assignment: *parts of TSF*], the TSF].

FPT_TST.1.2 The TSF shall provide authorized users the capability to verify the integrity of [selection: [assignment: *parts of TSF data*], TSF data].

FPT_TST.1.3 The TSF shall provide authorized users the capability to verify the integrity of [selection: [assignment: *parts of TSF*], TSF].

Application Note: This SFR is required only if "integrity failure via self-test" is selected in FPT_ARS_EXT.1. Self-tests typically include:

- Cryptographic module integrity checks (when crypto is implemented)
- File or image integrity validation
- Runtime environment verification
- Configuration integrity checks

12.3.2.1. Dependencies

Dependency	Resolution
No dependencies	N/A

12.4. Selection-Based Requirements Dependency Summary

Table 7. Selection-Based SFR Dependency Resolution

SFR	Trigger Condition	Dependencies	Resolution
FAU_STG_EX T.1	Audit export configured in FMT_SMF.1/Cloud	FAU_GEN.1, FAU_STG.1	Mandatory SFRs in this module

SFR	Trigger Condition	Dependencies	Resolution
FDP_DAR_EX T.1	Data-at-rest protection configured in FMT_SMF.1/Cloud	FCS_COP.1, FCS_CKM.1 (conditional)	DBMS Crypto Module (when encryption selected)
FDP_DIT_EX T.1	Data-in-transit protection configured in FMT_SMF.1/Cloud	FCS_TLSC_EXT.1, FCS_TLSS_EXT.1, FCS_HTTPS_EXT.1 (conditional)	DBMS Crypto Module (when TLS/HTTPS selected)
FPT_SBT_EX T.1	Secure boot selected in FPT_TUD_EXT.1	None	N/A
FPT_TST.1	"Integrity failure via self-test" selected in FPT_ARS_EXT.1	None	N/A

Chapter 13. Global Dependency Resolution Summary

This section provides a comprehensive view of all SFR dependencies and their resolutions for this PP-Module.

13.1. Dependency Resolution by Type

Table 8. Dependencies Satisfied by Module SFRs

Dependent SFR	Required Dependency	Satisfying SFR
FAU_SEL.1/Cloud	FAU_GEN.1	FAU_GEN.1 (this module)
FAU_SEL.1/Cloud	FMT_MTD.1	FMT_MTD_EXT.1 (this module)
FAU_STG.1	FAU_GEN.1	FAU_GEN.1 (this module)
FAU_STG_EXT.1	FAU_GEN.1, FAU_STG.1	FAU_GEN.1, FAU_STG.1 (this module)
FIA_UAU.1	FIA_UID.1	FIA_UID_EXT.1 (this module, hierarchical)
FIA_UID_EXT.1	FIA_UAU.1	FIA_UAU.1 (this module)
FMT_MOF.1	FMT_SMF.1, FMT_SMR.1	FMT_SMF.1/Cloud, FMT_SMR.1/Cloud (this module)
FMT_MTD_EXT.1	FMT_SMF.1, FMT_SMR.1	FMT_SMF.1/Cloud, FMT_SMR.1/Cloud (this module)
FMT_SMR.1/Cloud	FIA_UID.1	FIA_UID_EXT.1 (this module, hierarchical)

Table 9. Dependencies Satisfied by Operational Environment

Dependent SFR	Required Dependency	Resolution
FAU_GEN.1	FPT_STM.1	A.TIME_SYNCHRONIZATION (Trusted Platform provides time services)

Table 10. Dependencies Satisfied by External Modules (Conditional)

Dependent SFR	Required Dependency	Resolution
FDP_DAR_EXT.1 (when encryption selected)	FCS_COP.1, FCS_CKM.1	DBMS Crypto Module
FDP_DIT_EXT.1 (when TLS selected)	FCS_TLSC_EXT.1, FCS_TLSS_EXT.1	DBMS Crypto Module
FDP_DIT_EXT.1 (when HTTPS selected)	FCS_HTTPS_EXT.1	DBMS Crypto Module

13.2. Cryptographic SFR Exclusion Rationale

This PP-Module does not define SFRs from the FCS (Cryptographic Support) class. All cryptographic requirements are deferred to the DBMS Crypto Module for the following reasons:

1. **Separation of Concerns:** Cryptographic implementations require specialized evaluation activities and expertise. Separating cryptographic requirements into a dedicated module allows for focused evaluation.
2. **PP-Configuration Model:** The PP-Configuration approach allows flexible composition of security requirements. TOEs that do not implement cryptographic protections can claim conformance to this module without the Crypto Module.
3. **Reusability:** The DBMS Crypto Module can be shared across multiple PP-Modules (e.g., on-premises, cloud, DBaaS), ensuring consistent cryptographic requirements.
4. **FIPS/CAVP Alignment:** Separating cryptographic requirements facilitates alignment with FIPS 140-3 validation and CAVP testing requirements.

When cryptographic protections are selected in FDP_DAR_EXT.1 or FDP_DIT_EXT.1, the DBMS Crypto Module must be included in the PP-Configuration.

Appendix B: Optional Requirements

There are no Optional Requirements for this PP Module.

Appendix C: Extended Component Definitions

This appendix contains the definitions for the extended requirements that are used in the PP-Module, including those used in [Consistency Rationale](#) and [Selection-Based Requirements](#).

(Note: formatting conventions for selections and assignments in this chapter are those in [\[CC2\]](#).)

C.1. FIA: Identification and Authentication

C.1.1. FIA_UID_EXT: External Identity Integration

C.1.1.1. Family Behaviour

This family defines requirements for the TSF to support user identification through external cloud-native identity providers.

C.1.1.2. Component levelling

FIA_UID_EXT.1 External Identity Integration requires the TSF to identify users via external identity providers.

C.1.1.3. Management: FIA_UID_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of external identity provider connections. b) Mapping of external identities to internal roles.

C.1.1.4. Audit: FIA_UID_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Success or failure of external identity verification. b) Basic: Changes to identity provider configuration.

C.1.1.5. FIA_UID_EXT.1 External Identity Integration

Hierarchical to: No other components

Dependencies: FIA_UAU.1 Timing of Authentication

FIA_UID_EXT.1.1 The TSF shall allow [assignment: *list of TSF-mediated actions*] on behalf of the user to be performed before the user is identified.

FIA_UID_EXT.1.2 The TSF shall require each user to be successfully identified before allowing any other TSF-mediated actions on behalf of that user.

FIA_UID_EXT.1.3 The TSF shall provide user identification using external identity providers such as [selection: cloud IAM roles, federated directory users (e.g., LDAP/AD integrated with cloud IAM), service principals, [assignment: *other identity sources*]].

C.2. FMT: Security Management

C.2.1. FMT_MTD_EXT: Cloud Configuration State Protection

C.2.1.1. Family Behaviour

This family defines requirements for the TSF to manage and protect security-relevant data stored in cloud-native configuration and secret management systems.

C.2.1.2. Component levelling

FMT_MTD_EXT.1 Cloud Configuration State Protection requires the TSF to manage and protect TSF data in cloud-native systems.

C.2.1.3. Management: FMT_MTD_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of cloud secret management integrations. b) Rotation of cloud-managed credentials.

C.2.1.4. Audit: FMT_MTD_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Access to cloud-stored configuration data. b) Basic: Modification of cloud-stored secrets.

C.2.1.5. FMT_MTD_EXT.1 Cloud Configuration State Protection

Hierarchical to: No other components

Dependencies: FMT_SMF.1 Specification of Management Functions

FMT_MTD_EXT.1.1 The TSF shall restrict the ability to [selection: query, modify, delete, clear, [assignment: *other operations*]] the [assignment: *list of TSF data*] to [assignment: *the authorized identified roles*].

FMT_MTD_EXT.1.2 The TSF shall manage and protect TSF data stored in cloud-native configuration and secret management systems [assignment: *list of supported cloud secret management systems*].

C.3. FPT: Protection of the TSF

C.3.1. FPT_ARS_EXT: Availability and Resilience Signaling

C.3.1.1. Family Behaviour

This family defines requirements for the TSF to monitor its runtime environment and signal availability or resilience issues.

C.3.1.2. Component levelling

FPT_ARS_EXT.1 Availability and Resilience Signaling requires the TSF to monitor and alert on runtime environment issues.

C.3.1.3. Management: FPT_ARS_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of monitoring thresholds and alert recipients. b) Selection of monitored indicators.

C.3.1.4. Audit: FPT_ARS_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Detection of monitored conditions. b) Basic: Notification sent to administrators.

C.3.1.5. FPT_ARS_EXT.1 Availability and Resilience Signaling

Hierarchical to: No other components

Dependencies: No dependencies

FPT_ARS_EXT.1.1 The TSF shall monitor its runtime environment for indicators of [selection: configuration drift, resource unavailability, integrity failure via self-test, orchestration failure, [assignment: *other monitored conditions*]].

FPT_ARS_EXT.1.2 The TSF shall notify [assignment: *authorized administrator, orchestrator, or monitoring system*] upon detection of a monitored condition.

C.3.2. FPT_TUD_EXT: Trusted Update

C.3.2.1. Family Behaviour

This family defines requirements for the TSF to apply updates securely using trusted mechanisms.

C.3.2.2. Component levelling

FPT_TUD_EXT.1 Trusted Update requires the TSF to verify updates before installation.

C.3.2.3. Management: FPT_TUD_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of trusted update sources. b) Initiation of update procedures.

C.3.2.4. Audit: FPT_TUD_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Update verification success or failure. b) Basic: Update installation completion.

C.3.2.5. FPT_TUD_EXT.1 Trusted Update

Hierarchical to: No other components

Dependencies: No dependencies

FPT_TUD_EXT.1.1 The TSF shall provide authorized users the ability to query the current version of the TOE software/firmware.

FPT_TUD_EXT.1.2 The TSF shall provide a mechanism to apply updates using [assignment: *trusted update mechanisms*].

FPT_TUD_EXT.1.3 The TSF shall verify the authenticity and integrity of updates prior to installation using [assignment: *verification mechanisms*].

C.3.3. FPT_SBT_EXT: Secure Boot and Image Verification

C.3.3.1. Family Behaviour

This family defines requirements for the TSF to verify boot-time or deployment-time component integrity.

C.3.3.2. Component levelling

FPT_SBT_EXT.1 Secure Boot and Image Verification requires the TSF to verify deployment artifacts.

C.3.3.3. Management: FPT_SBT_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of trusted image sources or signing keys.

C.3.3.4. Audit: FPT_SBT_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Boot or image verification success or failure.

C.3.3.5. FPT_SBT_EXT.1 Secure Boot and Image Verification

Hierarchical to: No other components

Dependencies: No dependencies

FPT_SBT_EXT.1.1 The TSF shall verify the integrity of its boot-time or deployment-time components using [assignment: *verification mechanisms*].

FPT_SBT_EXT.1.2 The TSF shall verify the authenticity of its boot-time or deployment-time components by validating [selection: digital signatures from trusted signing authorities, cryptographic hashes against known-good values, [assignment: *other authenticity verification methods*]].

FPT_SBT_EXT.1.3 The TSF shall [selection: prevent boot/startup, alert administrators, [assignment: *other action*]] if integrity or authenticity verification fails.

C.4. FAU: Security Audit

C.4.1. FAU_STG_EXT: Audit Export

C.4.1.1. Family Behaviour

This family defines requirements for the TSF to export audit data to external systems.

C.4.1.2. Component levelling

FAU_STG_EXT.1 Audit Export requires the TSF to export audit data to cloud-native or SIEM services.

C.4.1.3. Management: FAU_STG_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of audit export destinations. b) Configuration of export format and frequency.

C.4.1.4. Audit: FAU_STG_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Audit export success or failure. b) Basic: Changes to export configuration.

C.4.1.5. FAU_STG_EXT.1 Audit Export

Hierarchical to: No other components

Dependencies: FAU_GEN.1 Audit data generation, FAU_STG.1 Protected audit trail storage

FAU_STG_EXT.1.1 The TSF shall support export of audit data to [assignment: *cloud-native logging or SIEM services*].

FAU_STG_EXT.1.2 The TSF shall export audit data using [selection: a trusted channel as specified in FTP_ITC.1, [assignment: *other secure export mechanism*]].

C.5. FDP: User Data Protection

C.5.1. FDP_DAR_EXT: Data-at-Rest Protection

C.5.1.1. Family Behaviour

This family defines requirements for the TSF to protect persistent storage of sensitive data.

C.5.1.2. Component levelling

FDP_DAR_EXT.1 Data-at-Rest Protection requires the TSF to protect stored data.

C.5.1.3. Management: FDP_DAR_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of encryption settings. b) Management of encryption keys.

C.5.1.4. Audit: FDP_DAR_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Enabling or disabling data-at-rest protection.

C.5.1.5. FDP_DAR_EXT.1 Data-at-Rest Protection

Hierarchical to: No other components

Dependencies: [FCS_COP.1 Cryptographic operation, FCS_CKM.1 Cryptographic key generation] or none

FDP_DAR_EXT.1.1 The TSF shall protect persistent storage of [assignment: *sensitive data types*] using [selection: encryption as defined in FCS_COP.1 and FCS_CKM.1 of the DBMS Crypto Module, disk encryption managed externally by the operational environment, none].

C.5.2. FDP_DIT_EXT: Data-in-Transit Protection

C.5.2.1. Family Behaviour

This family defines requirements for the TSF to protect data transmitted over networks.

C.5.2.2. Component levelling

FDP_DIT_EXT.1 Data-in-Transit Protection requires the TSF to protect data in transit.

C.5.2.3. Management: FDP_DIT_EXT.1

The following actions could be considered for the management functions in FMT:

a) Configuration of TLS/HTTPS settings. b) Management of certificates.

C.5.2.4. Audit: FDP_DIT_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP/ST:

a) Basic: Enabling or disabling data-in-transit protection. b) Basic: TLS/HTTPS connection failures.

C.5.2.5. FDP_DIT_EXT.1 Data-in-Transit Protection

Hierarchical to: No other components

Dependencies: [FCS_TLSC_EXT.1, FCS_TLSS_EXT.1, FCS_HTTPS_EXT.1] as defined in DBMS Crypto Module

FDP_DIT_EXT.1.1 The TSF shall protect data in transit between the TOE and external entities using [selection: TLS as defined in FCS_TLSC_EXT.1 and FCS_TLSS_EXT.1 of the DBMS Crypto Module, HTTPS as defined in FCS_HTTPS_EXT.1 of the DBMS Crypto Module, [assignment: *other cryptographic protocols*]].

Appendix D: Consistency Rationale

D.1. Consistency of Threats and Objectives

Table 11. Consistency Rationale for Threats

Threat	Consistency Rationale
T.ELASTIC_ENVIRONMENT	This threat is addressed by O.INTERNAL_RESOURCE_RESILIENCE (TSF detects disruptions, protects internal communications, and fails securely), OE.TRUSTED_PLATFORM (platform enforces isolation and resource protection), and OE.SECURE_ORCHESTRATION (deployment preserves security parameters).
T.MALICIOUS_TENANT_ADMIN	This threat is addressed by O.CLOUD_IDENTITY_INTEGRATION (proper authentication and role mapping), O.CLOUD_AUDIT_INTEGRATION (accountability through audit records), O.CONFIGURATION_PROTECTION (restricts configuration changes to authorized roles), and OE.SECURE_ORCHESTRATION (orchestration respects security policies).
T.UNAUTHORIZED_ACCESS	This threat is addressed by O.CLOUD_IDENTITY_INTEGRATION (authentication via cloud IAM), OE.TRUSTED_PLATFORM (tenant isolation), and OE.IDENTITY_PROVIDERS (accurate identity assertions).
T.INTEGRITY_FAILURE	This threat is addressed by O.CLOUD_DEPLOYMENT_INTEGRITY (verification of runtime state and artifacts), OE.SECURE_ORCHESTRATION (authentic and approved artifacts), and OE.TRUSTED_PLATFORM (protected resources).
T.AUDIT_FAILURE	This threat is addressed by O.CLOUD_AUDIT_INTEGRATION (audit generation and export) and OE.LOGGING_SERVICES (reliable audit storage pipeline).
T.SECRETS_EXPOSURE	This threat is addressed by O.CONFIGURATION_PROTECTION (protects security-critical configuration including credentials), OE.SECRETS_MANAGEMENT (secrets are authentic and protected from unauthorized disclosure), and OE.TRUSTED_PLATFORM (platform protects storage resources).
T.NETWORK_INTERCEPTION	This threat is addressed by O.INTERNAL_RESOURCE_RESILIENCE (protects internal communications from disclosure and modification), OE.TRUSTED_PLATFORM (network isolation between tenants), and selection-based FDP_DIT_EXT.1 (external network protection via DBMS Crypto Module).

D.2. Consistency of Assumptions and Objectives

Table 12. Consistency Rationale for Assumptions

Assumption	Consistency Rationale
A.TRUSTED_PLATFORM	This assumption is upheld by OE.TRUSTED_PLATFORM, which requires the platform to enforce tenant isolation, protect resources, and provide trusted identity assertions.
A.SECURE_DEPLOYMENT	This assumption is upheld by OE.SECURE_ORCHESTRATION, which requires orchestration systems to use authentic artifacts and preserve security parameters.
A.CLOUD_SERVICE_INTEGRITY	This assumption is upheld by OE.IDENTITY_PROVIDERS (correct identity assertions), OE.LOGGING_SERVICES (reliable audit services), and OE.SECRETS_MANAGEMENT (authentic secrets protected from disclosure).
A.AUTHORIZED_ADMINISTRATORS	This assumption is upheld by organizational security controls (personnel vetting, training, policies) that are prerequisite to the TOE's operational environment. The TOE provides accountability through O.CLOUD_AUDIT_INTEGRATION.
A.TIME_SYNCHRONIZATION	This assumption is upheld by OE.TRUSTED_PLATFORM, which requires the platform to provide trusted services to the TOE, including accurate time synchronization.

D.3. Consistency of Objectives and SFRs

Table 13. Consistency Rationale for TOE Objectives

TOE Objective	Consistency Rationale
O.CLOUD_IDENTITY_INTEGRATION	This objective is satisfied by FIA_UID_EXT.1 (external identity integration), FIA_UAU.1 (authentication enforcement), FMT_SMR.1/Cloud (tenant-scoped roles), FMT_SMF.1/Cloud (identity provider management), and FMT_MTD_EXT.1 (protection of cloud-stored credentials).
O.CLOUD_AUDIT_INTEGRATION	This objective is satisfied by FAU_GEN.1 (audit generation for cloud events), FAU_SEL.1/Cloud (tenant-aware audit selection), FAU_STG.1 (audit protection), FAU_STG_EXT.1 (audit export when selected), and FMT_SMF.1/Cloud (audit export configuration).
O.INTERNAL_RESOURCE_RESILIENCE	This objective is satisfied by FPT_ARS_EXT.1 (resilience monitoring and alerting), FPT_ITT.1 (protection of internal data transfers), FPT_FLS.1 (fail-secure from Base PP), and FPT_TST.1 (self-test when selected).
O.CLOUD_DEPLOYMENT_INTEGRITY	This objective is satisfied by FPT_TST.1 (self-test when selected), FPT_SBT_EXT.1 (secure boot when selected), and FPT_TUD_EXT.1 (trusted update verification).

TOE Objective	Consistency Rationale
O.CONFIGURATION_PROTECTION	This objective is satisfied by FMT_MOF.1 (restricts management of security functions to authorized roles), FMT_MTD_EXT.1 (protects TSF data in cloud configuration systems), FMT_SMR.1/Cloud (defines tenant-scoped roles), and FMT_SMF.1/Cloud (specifies management functions subject to protection).

Table 14. Consistency Rationale for Environmental Objectives

Environmental Objective	Consistency Rationale
OE.TRUSTED_PLATFORM	This objective is an operational environment requirement enforced by the cloud platform, not by TOE SFRs. The TOE relies on this objective as stated in A.TRUSTED_PLATFORM.
OE.SECURE_ORCHESTRATION	This objective is an operational environment requirement enforced by deployment systems, not by TOE SFRs. The TOE relies on this objective as stated in A.SECURE_DEPLOYMENT.
OE.LOGGING_SERVICES	This objective is an operational environment requirement enforced when FAU_STG_EXT.1 is selected, providing reliable external audit storage.
OE.IDENTITY_PROVIDERS	This objective is an operational environment requirement enforced when federated authentication is used, providing correct identity assertions consumed by FIA_UID_EXT.1.
OE.SECRETS_MANAGEMENT	This objective is an operational environment requirement enforced when the TOE integrates with cloud-native secret management systems as specified in FMT_MTD_EXT.1.

D.4. Consistency of Organizational Security Policies and Objectives

Table 15. Consistency Rationale for Organizational Security Policies

OSP	Consistency Rationale
P.AUDIT_RETENTION	This policy is supported by O.CLOUD_AUDIT_INTEGRATION, which requires the TOE to generate audit records and support export to external systems. FAU_STG_EXT.1 (when selected) enables organizations to implement retention policies using cloud-native storage. OE.LOGGING_SERVICES ensures the external storage is reliable.

OSP	Consistency Rationale
P.ACCESS_CONTROL_POLICY	This policy is supported by O.CLOUD_IDENTITY_INTEGRATION (role-based access control via cloud IAM), O.CONFIGURATION_PROTECTION (enforcement of role-based restrictions), FMT_SMR.1/Cloud (role definition), FMT_MOF.1 (restricts management functions to authorized roles), and O.CLOUD_AUDIT_INTEGRATION (logging of access control decisions).

Appendix E: SFR List

This table is provided as a reference of all SFRs included in this PP-Module.

The Type column has the following definitions:

Mandatory

The requirement must be included in the ST.

Selection-Based

The requirement must be included in the ST when selections in other SFRs trigger its inclusion, or when the TOE implements the specified functionality.

Optional

The requirement may be included in the ST at the ST author's discretion.

Table 16. Security Functional Requirements

Requirement Class	Requirement Component	Type
Security Audit (FAU)	FAU_GEN.1 Audit Data Generation	Mandatory
	FAU_SEL.1/Cloud Selective Audit (Cloud)	Mandatory
	FAU_STG.1 Protected Audit Trail Storage	Mandatory
	FAU_STG_EXT.1 Audit Export	Selection-Based
User Data Protection (FDP)	FDP_DAR_EXT.1 Data-at-Rest Protection	Selection-Based
	FDP_DIT_EXT.1 Data-in-Transit Protection	Selection-Based
Identification and Authentication (FIA)	FIA_UAU.1 Timing of Authentication	Mandatory
	FIA_UID_EXT.1 External Identity Integration	Mandatory
Security Management (FMT)	FMT_MOF.1 Management of Security Functions Behavior	Mandatory
	FMT_MTD_EXT.1 Cloud Configuration State Protection	Mandatory
	FMT_SMF.1/Cloud Specification of Management Functions (Cloud)	Mandatory
	FMT_SMR.1/Cloud Security Roles (Cloud)	Mandatory
Protection of the TSF (FPT)	FPT_ARS_EXT.1 Availability and Resilience Signaling	Mandatory
	FPT_ITT.1 Internal TSF Data Transfer Protection	Mandatory
	FPT_TUD_EXT.1 Trusted Update	Mandatory
	FPT_SBT_EXT.1 Secure Boot and Image Verification	Selection-Based
	FPT_TST.1 TSF Self-Test	Selection-Based

E.1. SFR Dependency Summary

The following table summarizes the dependencies between SFRs in this PP-Module:

Table 17. SFR Dependencies

SFR	Dependencies
FAU_GEN.1	FPT_STM.1 (satisfied by A.TIME_SYNCHRONIZATION)
FAU_SEL.1/Cloud	FAU_GEN.1, FMT_MTD.1 (satisfied by FMT_MTD_EXT.1)
FAU_STG.1	FAU_GEN.1
FAU_STG_EXT.1	FAU_GEN.1, FAU_STG.1
FIA_UAU.1	FIA_UID.1 (satisfied by FIA_UID_EXT.1)
FIA_UID_EXT.1	FIA_UAU.1
FMT_MOF.1	FMT_SMF.1, FMT_SMR.1
FMT_MTD_EXT.1	FMT_SMF.1
FMT_SMF.1/Cloud	None
FMT_SMR.1/Cloud	FIA_UID.1 (satisfied by FIA_UID_EXT.1)
FPT_ARS_EXT.1	None
FPT_ITT.1	None
FPT_TUD_EXT.1	None
FPT_SBT_EXT.1	None
FPT_TST.1	None
FDP_DAR_EXT.1	FCS_COP.1, FCS_CKM.1 (from DBMS Crypto Module, when encryption is selected)
FDP_DIT_EXT.1	FCS_TLSC_EXT.1, FCS_TLSS_EXT.1, FCS_HTTPS_EXT.1 (from DBMS Crypto Module, when TLS/HTTPS is selected)

Appendix F: Glossary

For the purpose of this PP-Module, the following terms and definitions apply. Terms defined in CC:2022 Part 1 [\[\[CC1\]\]](#) are included where they have particular relevance to this PP-Module.

Cloud IAM (Identity and Access Management)

Cloud-native services that manage user identities, authentication, and authorization within cloud environments. Examples include AWS IAM, Azure Active Directory, and Google Cloud IAM.

Data Encryption Key (DEK)

A cryptographic key used to encrypt and decrypt user data at rest. DEKs are typically protected by Key Encryption Keys (KEKs).

DBaaS (Database-as-a-Service)

A cloud service model where the cloud provider operates and manages the DBMS on behalf of tenants, providing database functionality as a fully managed service. This deployment model is explicitly out of scope for this PP-Module.

Elastic Environment

A cloud computing characteristic where resources (compute, storage, network) can be dynamically provisioned, scaled, or deprovisioned in response to changing workload demands. The TOE must maintain its security properties despite resource elasticity.

IaaS (Infrastructure-as-a-Service)

A cloud service model providing virtualized computing resources over the internet, including virtual machines, storage, and networking. The customer is responsible for managing the operating system, middleware, and applications deployed on IaaS resources.

Key Encryption Key (KEK)

A cryptographic key used to encrypt and protect other keys, particularly Data Encryption Keys (DEKs). KEKs are typically managed by key management services and may be stored in hardware security modules (HSMs).

Lift-and-Shift

A migration strategy where applications are moved to cloud infrastructure with minimal or no modifications to the application architecture. The TOE deployed via lift-and-shift is functionally equivalent to its on-premises version.

Orchestrator

A system or service responsible for automating the deployment, scaling, and management of containerized or virtualized workloads. Examples include Kubernetes, Docker Swarm, and cloud-native orchestration services. The TOE may be deployed by an orchestrator and must signal availability or integrity issues to the orchestrator.

PaaS (Platform-as-a-Service)

A cloud service model providing a platform for customers to develop, run, and manage applications without managing the underlying infrastructure. The cloud provider manages the

operating system, middleware, and runtime environment.

Service Principal

A non-human identity used by applications, services, or automated processes to authenticate to cloud services and access resources. Service principals are typically assigned roles and permissions similar to human users.

SIEM (Security Information and Event Management)

Systems that aggregate, correlate, and analyze security event data from various sources to provide centralized security monitoring, alerting, and compliance reporting.

Tenant

A customer or organization that uses cloud services within a shared infrastructure. In a multitenant cloud environment, isolation between tenants is enforced by the Trusted Platform.

Trusted Platform

The set of cloud infrastructure and services that enforce baseline security properties such as isolation, resource integrity, and identity attestation. The Trusted Platform is part of the TOE's operational environment and is not within the scope of evaluation. This concept aligns with the CCiTC architectural model.

Appendix G: Acronyms

Table 18. Acronyms

Acronym	Meaning
AD	Active Directory
AES	Advanced Encryption Standard
API	Application Programming Interface
CC	Common Criteria
CCiTC	Common Criteria in the Cloud Technical Community
cPP	collaborative Protection Profile
CSP	Cloud Service Provider
D@RE	Data-at-Rest Encryption
DBMS	Database Management System
DBaaS	Database-as-a-Service
DEK	Data Encryption Key
DIT	Data-in-Transit
EC	Exact Conformance
HSM	Hardware Security Module
HTTPS	Hypertext Transfer Protocol Secure
IAM	Identity and Access Management
IaaS	Infrastructure-as-a-Service
iTC	international Technical Community
KEK	Key Encryption Key
LDAP	Lightweight Directory Access Protocol
NIAP	National Information Assurance Partnership
OSP	Organizational Security Policy
PaaS	Platform-as-a-Service
PCL	Product Compliant List
PP	Protection Profile
SAR	Security Assurance Requirement
SFR	Security Functional Requirement
SIEM	Security Information and Event Management
SPD	Security Problem Definition
ST	Security Target
TLS	Transport Layer Security

Acronym	Meaning
TOE	Target of Evaluation
TSF	TOE Security Functionality
TSFI	TSF Interface
VM	Virtual Machine